

Code testing plan

This test plan is here to help validate the functionality and reliability of my Voting Analysis program. This includes testing user options and edge cases as well as seeing how the code handles unexpected inputs.

The test will cover the following:

- Analysing valid voters per party
- Listing constituencies
- Calculating gender percentages
- Listing MPs and parties
- Searching constituencies by region
- Displaying detailed constituency information
- Menu navigation and input test

Test 1.1 functional test

Test number ID	Option	Input	Expected output
T1.1	Analyse the number of valid votes per party	1	Correct count of valid votes per party
T1.2	List constituencies	2	All constituencies displayed
T1.3	Calculate % of male and female members	3	Male and female percentages displayed
T1.4	List MPs	4	MPs listed
T1.5	List parties	5	List of all parties
T1.6	List constituencies by region	6 and enter "Wales"	List constituencies in Wales
T1.7	Show constituency details	7 and enter "Newport East"	List details about Newport east

T1.1

```
--- Main Menu ---
Choose your option - use the number
Option Number    Option
1                Analyse the number of valid voters per party
2                List by constituency
3                Calculate % of Male and Female Members
4                List the MPS
5                List the parties
6                List constituencies by region
7                List constituency details
Enter the option number: 1
Lab: 17401690 valid votes
SNP: 398434 valid votes
Con: 5815132 valid votes
RUK: 206592 valid votes
LD: 3608321 valid votes
DUP: 209925 valid votes
SF: 311596 valid votes
SDLP: 81849 valid votes
Ind: 248572 valid votes
Green: 195162 valid votes
PC: 164929 valid votes
Spk: 33964 valid votes
APNI: 49081 valid votes
TUV: 41185 valid votes
UUP: 42908 valid votes
Do you want to try again (Y) or return to the main menu (M)? █
```

T1.2

```

--- Main Menu ---
Choose your option - use the number
Option Number  Option
1              Analyse the number of valid voters per party
2              List by constituency
3              Calculate % of Male and Female Members
4              List the MPS
5              List the parties
6              List constituencies by region
7              List constituency details
Enter the option number: 2
Aberafan Maesteg - Lab (Wales)
Aberdeen North - SNP (Scotland)
Aberdeen South - SNP (Scotland)
Aberdeenshire North and Moray East - SNP (Scotland)
Airdrie and Shotts - Lab (Scotland)
Aldershot - Lab (South East)
Aldridge-Brownhills - Con (West Midlands)
Alloa and Grangemouth - Lab (Scotland)
Altrincham and Sale West - Lab (North West)
Alyn and Deeside - Lab (Wales)
Amber Valley - Lab (East Midlands)
Angus and Perthshire Glens - SNP (Scotland)
Arbroath and Broughty Ferry - SNP (Scotland)
Argyll Bute and South Lochaber - SNP (Scotland)
Arundel and South Downs - Con (South East)
Ashfield - RUK (East Midlands)
Ashford - Lab (South East)
Ashton-under-Lyne - Lab (North West)
Aylesbury - Lab (South East)
Ayr Carrick and Cumnock - Lab (Scotland)
Banbury - Lab (South East)
Bangor Aberconwy - Lab (Wales)
Barking - Lab (London)
Barnsley North - Lab (Yorkshire and The Humber)
Barnsley South - Lab (Yorkshire and The Humber)
Barrow and Furness - Lab (North West)
Basildon and Billericay - Con (East of England)
Basingstoke - Lab (South East)
Bassetlaw - Lab (East Midlands)
Bath - LD (South West)
Bathgate and Linlithgow - Lab (Scotland)
Battersea - Lab (London)
Beaconsfield - Con (South East)
Beckenham and Penge - Lab (London)
Bedford - Lab (East of England)
Belfast East - DUP (Northern Ireland)
Belfast North - SF (Northern Ireland)
Belfast South and Mid Down - SDLP (Northern Ireland)
Belfast West - SF (Northern Ireland)

```

```

--- Main Menu ---
Choose your option - use the number
Option Number  Option
1              Analyse the number of valid voters per party
2              List by constituency
3              Calculate % of Male and Female Members
4              List the MPS
5              List the parties
6              List constituencies by region
7              List constituency details
Enter the option number: 3
Male Members: 387 (59.54%)
Female Members: 263 (40.46%)
Do you want to try again (Y) or return to the main menu (M)? █

```

T1.4

```

--- Main Menu ---
Choose your option - use the number
Option Number  Option
1              Analyse the number of valid voters per party
2              List by constituency
3              Calculate % of Male and Female Members
4              List the MPs
5              List the parties
6              List constituencies by region
7              List constituency details
8              Exit
Enter the option number: 4

Number of MPs in each party:
Lab: 411 MPs
Con: 121 MPs
LD: 72 MPs
RUK: 5 MPs
Green: 4 MPs
IND: 0 MPs
SNP: 9 MPs
PC: 4 MPs
DUP: 5 MPs
SF: 7 MPs
SDLP: 2 MPs
UUP: 1 MPs
APNI: 1 MPs
Do you want to try again (Y) or return to the main menu (M)? █

```

T1.5

```
--- Main Menu ---
Choose your option - use the number
Option Number  Option
1      Analyse the number of valid voters per party
2      List by constituency
3      Calculate % of Male and Female Members
4      List the MPS
5      List the parties
6      List constituencies by region
7      List constituency details
Enter the option number: 5
List of Parties: ['Lab', 'Con', 'LD', 'RUK', 'Green', 'IND', 'SNP', 'PC', 'DUP', 'SF', 'SDLP', 'UUP', 'APNI']
Do you want to try again (Y) or return to the main menu (M)? █
```

T.6

--- Main Menu ---

Choose your option - use the number

Option Number Option

- 1 Analyse the number of valid voters per party
- 2 List by constituency
- 3 Calculate % of Male and Female Members
- 4 List the MPS
- 5 List the parties
- 6 List constituencies by region
- 7 List constituency details

Enter the option number: 6

Enter the region name: wwlee

Constituencies in wwlee:

No constituencies found in that region.

Do you want to try again (Y) or return to the main menu (M)? Y

Enter the region name: Wales

Constituencies in Wales:

Aberafan Maesteg - Lab (Wales)
Alyn and Deeside - Lab (Wales)
Bangor Aberconwy - Lab (Wales)
Blaenau Gwent and Rhymney - Lab (Wales)
Brecon Radnor and Cwm Tawe - LD (Wales)
Bridgend - Lab (Wales)
Caerfyrddin - PC (Wales)
Caerphilly - Lab (Wales)
Cardiff East - Lab (Wales)
Cardiff North - Lab (Wales)
Cardiff South and Penarth - Lab (Wales)
Cardiff West - Lab (Wales)
Ceredigion Preseli - PC (Wales)
Clwyd East - Lab (Wales)
Clwyd North - Lab (Wales)
Dwyfor Meirionnydd - PC (Wales)
Gower - Lab (Wales)
Llanelli - Lab (Wales)
Merthyr Tydfil and Aberdare - Lab (Wales)
Mid and South Pembrokeshire - Lab (Wales)
Monmouthshire - Lab (Wales)
Montgomeryshire and Glyndŵr - Lab (Wales)
Neath and Swansea East - Lab (Wales)
Newport East - Lab (Wales)
Newport West and Islwyn - Lab (Wales)
Pontypridd - Lab (Wales)
Rhondda and Ogmore - Lab (Wales)
Swansea West - Lab (Wales)
Torfaen - Lab (Wales)
Vale of Glamorgan - Lab (Wales)
Wrexham - Lab (Wales)
Ynys Mŷn - PC (Wales)

T1.7

```
--- Main Menu ---
Choose your option - use the number
Option Number  Option
1              Analyse the number of valid voters per party
2              List by constituency
3              Calculate % of Male and Female Members
4              List the MPS
5              List the parties
6              List constituencies by region
7              List constituency details
Enter the option number: 7
Enter the constituency name: neportest
Constituency not found.
Do you want to try again (Y) or return to the main menu (M)? Y
Enter the constituency name: newport east
Newport East - Lab (Wales)
Details for newport east:
Region: County
Country: Wales
Party: Lab
Valid Votes: 38531
Majority: 9009

--- Main Menu ---
Choose your option - use the number
Option Number  Option
1              Analyse the number of valid voters per party
2              List by constituency
3              Calculate % of Male and Female Members
4              List the MPS
5              List the parties
6              List constituencies by region
7              List constituency details
Enter the option number: █
```

Test 2.1 Negative tests

Test number ID	Feature	Input	Expected output
T2.1	Invalid menu option	"8"	"Invalid option. Please try again". Re prints main menu
T2.2	Search invalid region	"Ireland"	When invalid region entered it should prompt for user to try again.
T2.3	Search Invalid constituency	"Dublin"	When invalid constituency is

			entered it should prompt user to retry.
--	--	--	---

T2.1

```

--- Main Menu ---
Choose your option - use the number
Option Number  Option
1              Analyse the number of valid voters per party
2              List by constituency
3              Calculate % of Male and Female Members
4              List the MPS
5              List the parties
6              List constituencies by region
7              List constituency details
Enter the option number: 8
Invalid option. Please try again.

```

```

--- Main Menu ---
Choose your option - use the number
Option Number  Option
1              Analyse the number of valid voters per party
2              List by constituency
3              Calculate % of Male and Female Members
4              List the MPS
5              List the parties
6              List constituencies by region
7              List constituency details
Enter the option number: 

```

T2.2


```

---- Main Menu ----
Choose your option - use the number
Option Number    Option
1                Analyse the number of valid voters per party
2                List by constituency
3                Calculate % of Male and Female Members
4                List the MPS
5                List the parties
6                List constituencies by region
7                List constituency details
Enter the option number: 6
Enter the region name: Ireland
Constituencies in Ireland:
No constituencies found in that region.
Do you want to try again (Y) or return to the main menu (M)? Y
Enter the region name: 

```

T2.3

```

---- Main Menu ----
Choose your option - use the number
Option Number    Option
1                Analyse the number of valid voters per party
2                List by constituency
3                Calculate % of Male and Female Members
4                List the MPS
5                List the parties
6                List constituencies by region
7                List constituency details
Enter the option number: 7
Enter the constituency name: Dublin
Constituency not found.
Do you want to try again (Y) or return to the main menu (M)? Y
Enter the constituency name: 

```

Test 3.1 edge cases

Test number ID	Feature	Input	Expected output
T3.1	Empty constituency file	Run program with no data	Appropriate error message
T3.2	Mixed case input	Input region constituency in mixed case. E.g wAIEs	Case mismatching autocorrection. Info should still be outputted.

3.1

```

/usr/local/bin/python3 /Users/mathewmadzibanyika/Documents/GitHub/VotingAnalysisProject/vap2.py
mathewmadzibanyika@Apple-Macbook-Air-13-in-M2 VotingAnalysisProject % /usr/local/bin/python3 /Users/mathewmadzibanyika/Documents/GitHub/VotingAnalysisProject/vap2.py
Traceback (most recent call last):
  File "/Users/mathewmadzibanyika/Documents/GitHub/VotingAnalysisProject/vap2.py", line 75, in <module>
    with open('/Users/mathewmadzibanyika/Documents/GitHub/VotingAnalysisProject/EditedDat.csv', newline='') as csvfile:
FileNotFoundError: [Errno 2] No such file or directory: '/Users/mathewmadzibanyika/Documents/GitHub/VotingAnalysisProject/EditedDat.csv'
mathewmadzibanyika@Apple-Macbook-Air-13-in-M2 VotingAnalysisProject %

```

3.2

```

--- Main Menu ---
Choose your option - use the number
Option Number    Option
1                Analyse the number of valid voters per party
2                List by constituency
3                Calculate % of Male and Female Members
4                List the MPS
5                List the parties
6                List constituencies by region
7                List constituency details
Enter the option number: 6
Enter the region name: wALes
Constituencies in wALes:
Aberafan Maesteg - Lab (Wales)
Alyn and Deeside - Lab (Wales)
Bangor Aberconwy - Lab (Wales)
Blaenau Gwent and Rhymney - Lab (Wales)
Brecon Radnor and Cwm Tawe - LD (Wales)
Bridgend - Lab (Wales)
Caerfyrddin - PC (Wales)
Caerphilly - Lab (Wales)
Cardiff East - Lab (Wales)
Cardiff North - Lab (Wales)
Cardiff South and Penarth - Lab (Wales)
Cardiff West - Lab (Wales)
Ceredigion Preseli - PC (Wales)
Clwyd East - Lab (Wales)
Clwyd North - Lab (Wales)
Dwyfor Meirionnydd - PC (Wales)
Gower - Lab (Wales)
Llanelli - Lab (Wales)
Merthyr Tydfil and Aberdare - Lab (Wales)
Mid and South Pembrokeshire - Lab (Wales)
Monmouthshire - Lab (Wales)
Montgomeryshire and Glyndŵr - Lab (Wales)
Neath and Swansea East - Lab (Wales)
Newport East - Lab (Wales)
Newport West and Islwyn - Lab (Wales)
Pontypridd - Lab (Wales)
Rhondda and Ogmore - Lab (Wales)
Swansea West - Lab (Wales)
Torfaen - Lab (Wales)
Vale of Glamorgan - Lab (Wales)
Wrexham - Lab (Wales)
Ynys Mŷn - PC (Wales)
Do you want to try again (Y) or return to the main menu (M)? 

```